

Wazuh Installation & FIM Monitoring

1. Objective

The objective of this task was to install Wazuh, collect logs, configure File Integrity Monitoring (FIM), and generate alerts when monitored files are modified.

2. Environment Details

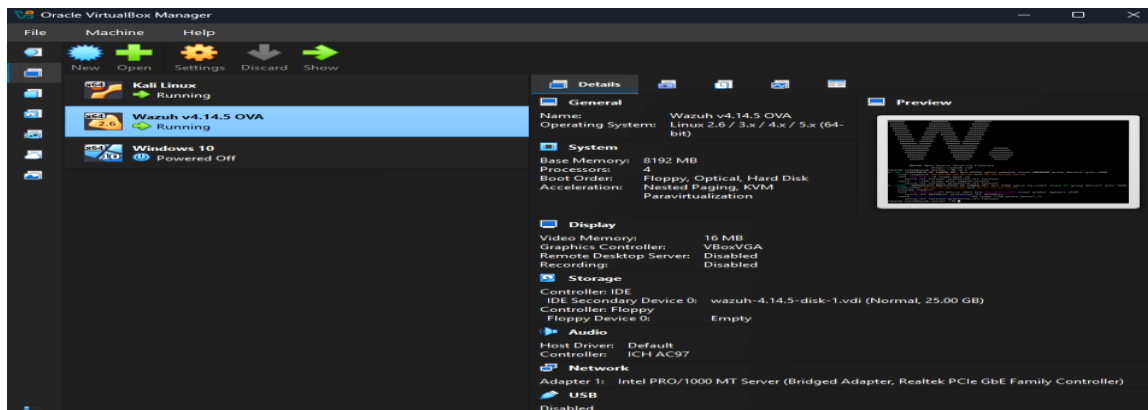
OS: Kali Linux 2026.1

Wazuh Version: v4.14.5 OVA

Virtual Machine: Oracle Virtual Box

3. Wazuh Installation

I installed the Wazuh SIEM platform on Oracle VirtualBox as a separate virtual machine (VM) environment.

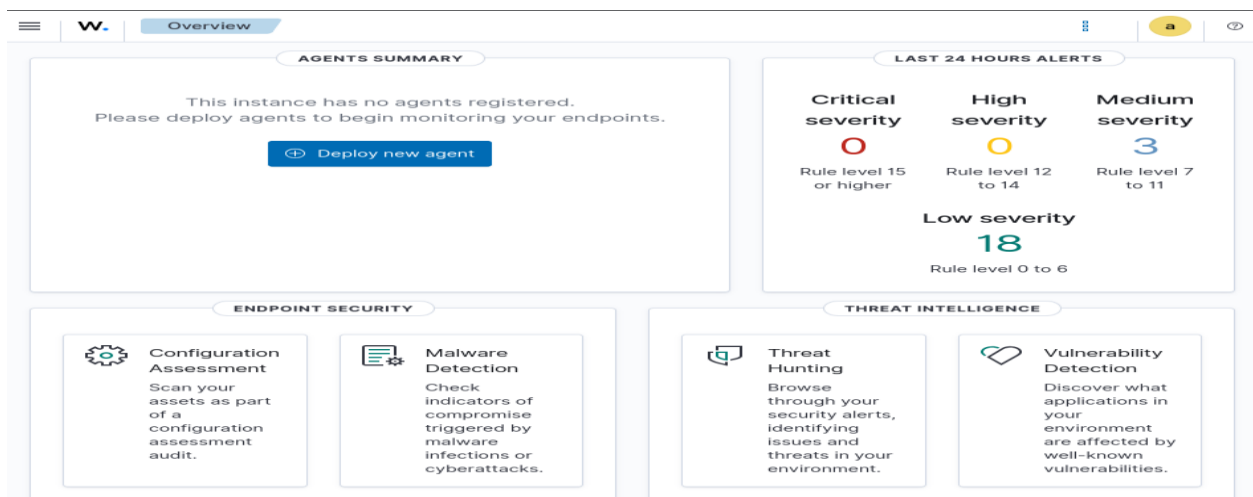


```
WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0d:10:83 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 192.168.1.5/24 metric 1024 brd 192.168.1.255 scope global dynamic eth0
        valid_lft 86268sec preferred_lft 86268sec
    inet6 fe80::a00:27ff:fe0d:1083/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
```

4. Wazuh Setup

Accessing the Wazuh Dashboard

- Opened the wazuh web interface through the browser using IP of wazuh server.
- Logged in using the login credentials.
- Verified successful access to the Dashboard.



Agent Installation and Registration

- Installed the wazuh agent on the monitored system

- Configured the manager ip address.
- Registered the agent with the wazuh manager.
- Confirmed the agent status as Active.

Agents (1)

+

Deploy new agent

↺

Refresh

📄

Export formatted

More

▼

⚙️

status=active

WQL

☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	001	Kali_Linux	192.168.1.7	default	 Kali GNU/Linux 2026.1	node01	v4.14.5	ⓘ	👁 ⋮

Rows per page: 10

▼

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5. Log Collection Verification

- Confirmed that the wazuh agent was forwarding logs to the manager.
- Verified logs appearing in the dashboard under Security Events.
- Ensured real-time monitoring was functioning properly.

```

(root@kali)-[/home/user/Desktop]
# exit

(user@kali)-[~]
$ sudo su
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
1Sorry, try again.
[sudo] password for user:
sudo: 3 incorrect password attempts

(user@kali)-[~]
$ sudo su
[sudo] password for user:
(root@kali)-[/home/user]
#

```

Search

DQL

Last 24 hours

Show dates

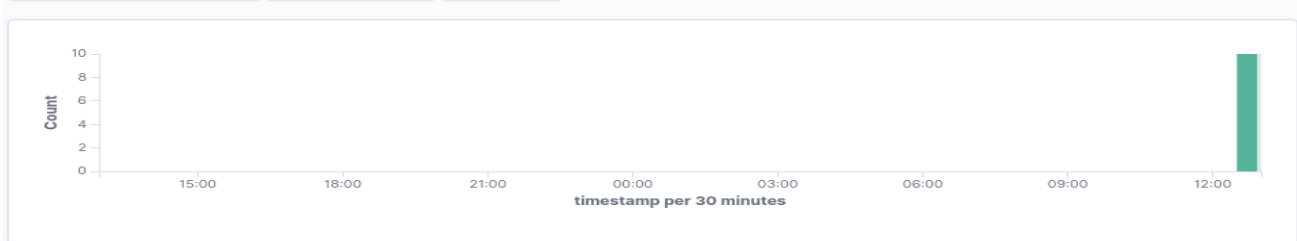
Refresh

manager.name: wazuh-server

rule.mitre.id: exists

agent.id: 001

Add filter



10 hits

Jun 23, 2026 @ 12:57:21.428 - Jun 24, 2026 @ 12:57:21.428

Export Formatted

Reset view

623 available fields

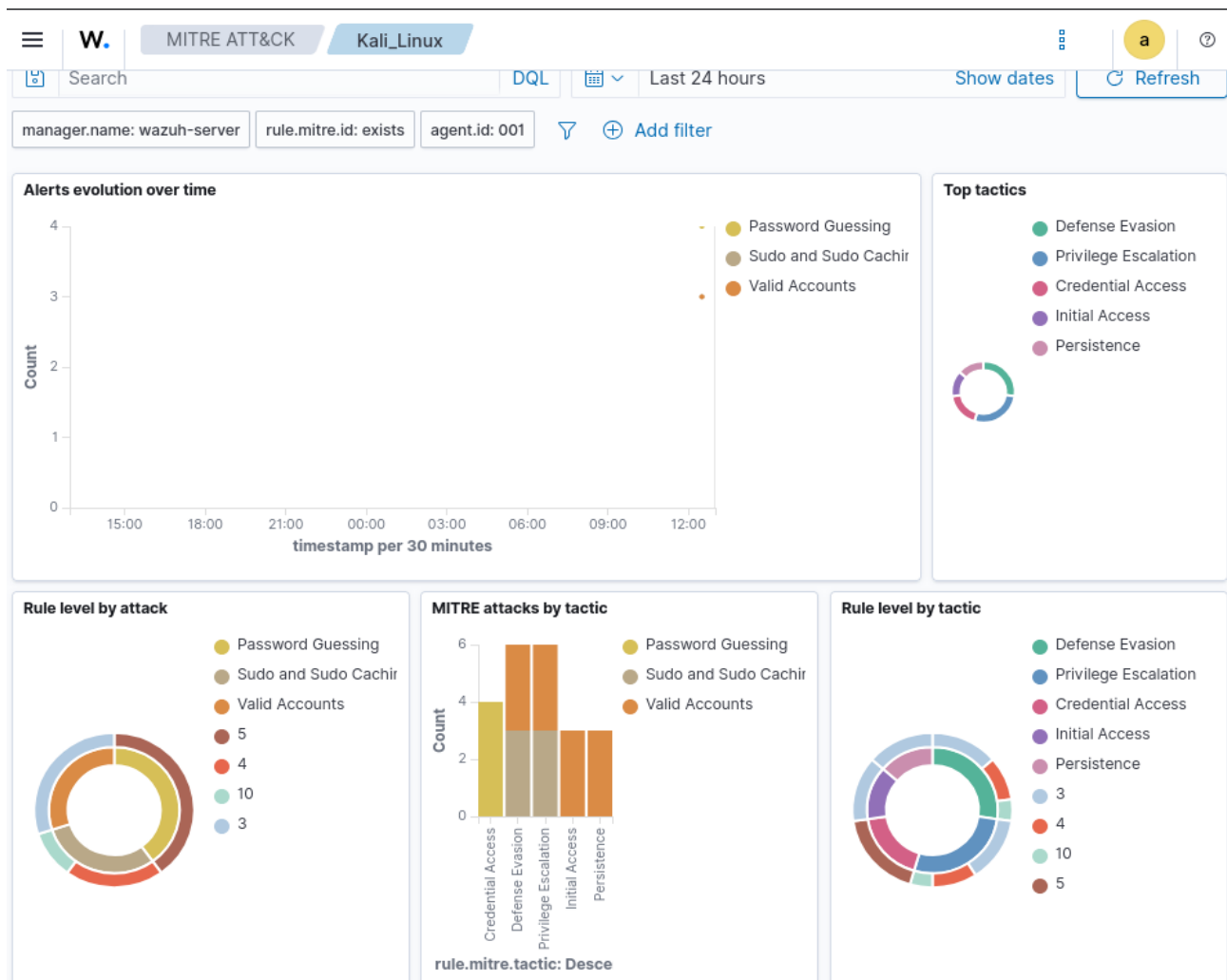
Columns

Density

1 fields sorted

Full screen

	timestamp	agent.name	rule.mitre.id	rule.mitre.tactic
	Jun 24, 2026 @ 12:55:15.1...	Kali_Linux	T1078	Defense Evasion, Persistence, Privileg..
	Jun 24, 2026 @ 12:5...	Kali_Linux	T1078	Defense Evasion, Persistence, Privileg..
	Jun 24, 2026 @ 12:55:15.1...	Kali_Linux	T1548.003	Privilege Escalation, Defense Evasion
	Jun 24, 2026 @ 12:55:08.9...	Kali_Linux	T1548.003	Privilege Escalation, Defense Evasion
	Jun 24, 2026 @ 12:55:04.9...	Kali_Linux	T1110.001	Credential Access
	Jun 24, 2026 @ 12:55:02.9...	Kali_Linux	T1110.001	Credential Access
	Jun 24, 2026 @ 12:54:58.9...	Kali_Linux	T1110.001	Credential Access



6. FIM Configuration

- Enables FIM (Syscheck) module.
- Configured directory monitoring.
- Selected specific directory for monitoring.
- Restarted wazuh services.

```

root@kali: /home/user
Session Actions Edit View Help
GNU nano 8.7.1 /var/ossec/etc/ossec.conf

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>
  <directories check_all="yes" report_changes="yes" realtime="yes">/home/u>
  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>

```

7. FIM Testing

- Created a File.
- Modified that File.

```

Session Actions Edit View Help

(root@kali)-[/home/user/Desktop/FIM]
# touch test.txt

(root@kali)-[/home/user/Desktop/FIM]
# nano test.txt

(root@kali)-[/home/user/Desktop/FIM]
#

```

- Got both creation & modification alerts.

Document Details	View surrounding documents	View single document
† agent.name	Kali_Linux	
† decoder.name	syscheck_new_entry	
† full_log	File '/home/user/Desktop/FIM/test.txt' added Mode: realtime	

Document Details	View surrounding documents	View single document
† agent.name	Kali_Linux	
† decoder.name	syscheck_integrity_changed	
† full_log	File '/home/user/Desktop/FIM/test.txt' modified Mode: realtime Changed attributes: size,m time,md5,sha1,sha256 Size changed from '0' to '14'	

8. Conclusion

In this task, Wazuh was successfully deployed and configured to collect and monitor system logs in real time. The wazuh dashboard was verified to be functioning correctly, and the connected agent successfully forwarded the security alerts.

FIM was configured to monitor selected directories and enabling the detection of integration in file. Various tasks are performed to generate file related events, and the corresponding alerts are successfully observed in the wazuh dashboard. This demonstrates the effectiveness of wazuh as a SIEM solution for centralized logs management, security monitoring, and threat detection.